SI Core

Quick Start Guide | Sunsynk data logger + Solaire Intelligence Pi

QUICK START

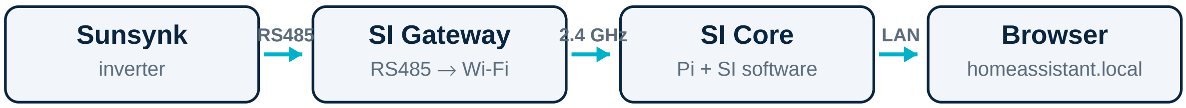
Rev. 1.0

WELCOME TO SOLAIRE INTELLIGENCE

Let’s get your system online.

SI Core is your Solaire Intelligence brain — a Raspberry Pi pre-flashed with our platform, paired with one or more SI Gateways that log data from your **Sunsynk** inverter(s) over RS485. Get the hardware in place and we take it from there — remotely.

HOW IT FITS TOGETHER



IN THE SI CORE BOX

Brain

- **SI Core** — Raspberry Pi (pre-configured)
- SSD storage (pre-loaded with SI software)
- USB-C power supply
- Ethernet patch cable

IN EACH GATEWAY BOX

Bridge

- **SI Gateway** — one per inverter
- RS485 → RJ45 cable
- Gateway power supply
- (multi-inverter sites: one kit per inverter)

WHAT YOU’LL NEED

Before you start

- A qualified solar installer to open the inverter
- Your home Wi-Fi name and password (2.4 GHz)
- A network cable run to your router/switch
- A phone or laptop (for Gateway pairing)

AT A GLANCE

The 6 steps ahead

1	Unpack & identify the kit
2	Install the SI Gateway at the inverter
3	Connect the Gateway to your Wi-Fi
4	Plug in and boot the SI Core (Raspberry Pi)
5	Open the system in your browser
6	We take it from here

1 Unpack & identify the kit

Lay everything out on a clean surface and check it against the lists on page 1. For multi-inverter sites, the SI Core box is shared by all inverters; each inverter gets its own SI Gateway kit.

- **SI Core** — small box with a Raspberry Pi and USB-C PSU inside.
- **SI Gateway** — smaller box; contains the RS485 bridge, an RJ45 cable, and a plug-pack.
- Check the **label on the Pi itself** — it carries the **unique Home Assistant username and password** for this SI Core. You'll need them in Step 5.

2 Install the SI Gateway at the inverter

Qualified installer only — isolate before opening

A qualified solar installer should perform this step. Switch **off the DC isolator** and the AC breaker feeding the inverter **before** opening the front panel. Sunsynk inverters carry lethal voltages internally even when the display is off.

With the inverter safely isolated, wire the Gateway into the inverter's RS485 port:

- Open the front cover of the Sunsynk inverter.
- Feed the **RS485** → **RJ45** cable through the gland at the bottom of the inverter.
- Plug the RJ45 into the port labelled **RS485** on the inverter's comms board.
- Route the other end of the cable to the Gateway and plug it in.
- Plug the Gateway's power supply into a nearby 230 V outlet.
- Close the inverter panel and restore power.

Multi-inverter sites

Install one Gateway **per inverter**. Each Gateway is paired to its own inverter over RS485 and reports independently to the SI Core. Label each Gateway ("Inv 1", "Inv 2"…) so we can identify them during remote configuration.

3 Connect the Gateway to your Wi-Fi

On first boot, the Gateway puts up its own temporary Wi-Fi hotspot. Use a phone or laptop to tell it which home Wi-Fi network to join.

Hotspot network (SSID)	`SI Gateway`
Band	2.4 GHz only
Same network as	the SI Core (Raspberry Pi) — see Step 4

- On your phone, open Wi-Fi settings and connect to **SI Gateway**.
- Wait a few seconds — a setup screen should appear automatically.
- Pick your home Wi-Fi network from the list and enter its password.
- Save. The Gateway restarts and joins your network (about 30 seconds).

Two Wi-Fi rules

1. Only **2.4 GHz** networks are supported. If your router uses one combined SSID for 2.4 & 5 GHz, that's fine. 2. The Gateway **must** be on the same LAN as the SI Core. Don't put the Gateway on a guest network.

The Gateway has a small status LED visible on the case. It tells you at a glance where the Gateway is in the handshake with your inverter and SI Core.

	Colour	Means	What to do
●	**Yellow**	Waiting for pairing	The Gateway has booted but isn't yet on a Wi-Fi network. Follow Step 3.
●	**Blue**	Wi-Fi connected — waiting for inverter signal	Gateway is on your Wi-Fi and can reach the SI Core, but it hasn't heard the inverter on RS485 yet. Check the cable and that the inverter is powered.
●	**Green**	Modbus communicating	All good — the Gateway is reading live data from the Sunsynk over RS485 and forwarding it to the SI Core.
●	**Red**	Communication lost	The Gateway has stopped hearing the inverter or the SI Core. Check RS485 wiring, then Wi-Fi.

The LED pulses slowly between **red** and **green** — that's the operating **heartbeat**. A steady green (or a slow red/green pulse) means the Gateway is alive and talking to the inverter.

4 Plug in and boot the SI Core

The SI Core is a Raspberry Pi already flashed with the Solaire Intelligence image and connected to its SSD. You just need to give it power and a network connection.

- Connect an Ethernet patch cable from the Pi to your **router or network switch**.
- Plug the USB-C power supply into the Pi and into a wall outlet.
- Wait **2–3 minutes** for the first boot to complete.

> **Wired Ethernet is required**
>
> The SI Core is designed to run over wired Ethernet for reliability. It must be on the **same LAN** as the Gateway so the two can talk to each other.

5 Open the system in your browser

Once the Pi has finished booting, you can reach the dashboard from any computer, phone, or tablet on the same network.

Open a browser and go to	`http://homeassistant.local:8123`
Log in with	the **username and password printed on the label on your Pi**. Each SI Core ships with its own unique credentials — they're not shared between units.

> **Can't reach homeassistant.local?**
>
> Some older Windows PCs and certain routers don't resolve **.local** names. As a fallback, look in your router's DHCP client list for a device called **homeassistant** or **raspberrypi** and use its IP address instead, e.g. **http://192.168.1.42:8123**.

6 We take it from here

With the hardware powered and on the network, **no further action is required from you**. The SI team connects to your system remotely and handles the rest.

What happens	Detail
Automatic inverter discovery	Your SI Core will detect the Gateway(s) on the LAN and start receiving live data from each Sunsynk inverter.
Remote configuration	Our team logs in remotely, binds each Gateway to the correct inverter, labels all the data points (battery SOC, PV, grid, load, etc.) and sets up your dashboard.
Optimisation & handover	We review the first day of data, tune alert thresholds, and confirm everything is reporting cleanly. You'll get a short welcome email with your dashboard link and a tour.

TROUBLESHOOTING

If something doesn't look right

Symptom	Try this
Gateway hotspot ("SI Gateway") doesn't show up on my phone	Check the Gateway's power supply is plugged in and the RS485 cable is seated. Unplug the Gateway for 10 seconds and plug it back in. Toggle your phone's Wi-Fi off and on.
I entered my Wi-Fi password but the Gateway won't join	Double-check the password for typos. Confirm the network is **2.4 GHz** (not 5 GHz only) and that it's broadcasting its SSID.
No inverter data after 10 minutes	Confirm the RS485 cable is in the correct port on the Sunsynk (labelled **RS485**, not CAN or BMS). Make sure the Gateway and the SI Core are on the **same** LAN / subnet.
Multiple inverters — only one is showing up	Each inverter needs its own Gateway. Confirm each Gateway was paired to Wi-Fi in Step 3 — and labelled so our team can match it to the right inverter during remote setup.

Welcome to the SI family

You're now part of a smarter, more efficient energy system. If anything looks off during the first 24 hours, drop us a line — we'll check it remotely.

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Web
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